

## CHAING RAI ARGOMET, Thailand

WMO: 483040

Lat: 19.8719N

Lon: 99.7793E

Elev: 1316

StdP: 14.01

Time Zone: 7.00 (E07)

Period: 09-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | 51.7       | 54.9 | 48.0                          | 52.0 | 60.7 | 51.2 | 58.6 | 63.2 | 9.8                   | 63.6 | 7.6 | 71.3 | 0.1                      | N/A  | 0.257 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 6             | 14.0                   | 98.1            | 75.7 | 95.6 | 75.9 | 93.4 | 75.8 | 80.9                | 89.1 | 80.1 | 88.3 | 79.5 | 87.7 | 2.3                  | N/A  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 78.9                            | 157.6 | 85.4 | 78.1 | 153.4 | 84.5 | 77.4 | 149.8 | 83.6 | 45.9          | 89.5 | 45.1 | 88.5 | 44.4 | 87.7 | 84.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|
| 1%                | 2.5% | 5%  | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |
| (a)               | (b)  | (c) | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |
| 7.7               | 6.4  | 5.1 | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |
|                   |      |     | 46.6                       | 101.5 | 3.6                | 3.0 | 44.0                                               | 103.6 | 41.9       | 105.4 | 39.9       | 107.0 | 37.3       | 109.2 |
|                   |      |     | 46.7                       | 82.7  | 3.9                | 0.8 | 43.9                                               | 83.3  | 41.6       | 83.8  | 39.4       | 84.3  | 36.6       | 84.9  |
|                   |      |     |                            | DB    |                    |     |                                                    |       |            |       |            |       |            |       |
|                   |      |     |                            | WB    |                    |     |                                                    |       |            |       |            |       |            |       |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 78.0  | 69.8  | 73.2  | 77.3  | 81.7  | 82.9  | 82.6  | 81.4  | 80.8  | 80.8  | 79.0  | 75.6  | 70.3 | (6)  |
| (7)  |                                                                   | DBStd       | 5.48  | 3.77  | 3.10  | 3.51  | 3.61  | 3.39  | 2.18  | 2.44  | 1.99  | 1.95  | 2.51  | 3.36  | 4.33 | (7)  |
| (8)  |                                                                   | HDD50       | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (8)  |
| (9)  |                                                                   | HDD65       | 24    | 10    | 1     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 12   | (9)  |
| (10) |                                                                   | CDD50       | 10203 | 613   | 649   | 845   | 951   | 1019  | 977   | 974   | 954   | 924   | 900   | 768   | 629  | (10) |
| (11) |                                                                   | CDD65       | 4752  | 158   | 230   | 381   | 501   | 554   | 527   | 509   | 489   | 474   | 435   | 318   | 176  | (11) |
| (12) |                                                                   | CDH74       | 42622 | 1049  | 2117  | 3664  | 5241  | 5687  | 5362  | 4736  | 4169  | 4067  | 3310  | 2175  | 1045 | (12) |
| (13) |                                                                   | CDH80       | 16837 | 212   | 853   | 1831  | 2797  | 2706  | 2100  | 1618  | 1273  | 1371  | 1143  | 711   | 222  | (13) |
| (14) | Wind                                                              | WSAvg       | 0.6   | 0.4   | 0.5   | 0.5   | 0.7   | 0.8   | 0.9   | 0.8   | 0.5   | 0.4   | 0.5   | 0.4   | 0.4  | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 69.4  | 0.5   | 0.2   | 1.1   | 2.8   | 8.0   | 9.0   | 11.7  | 16.9  | 10.2  | 5.6   | 2.4   | 1.2  | (15) |
| (16) |                                                                   | PrecMax     | 86.9  | 3.1   | 0.9   | 4.6   | 9.4   | 13.3  | 16.6  | 19.1  | 25.2  | 17.8  | 11.0  | 6.7   | 6.3  | (16) |
| (17) |                                                                   | PrecMin     | 48.6  | 0.0   | 0.0   | 0.0   | 0.1   | 2.5   | 4.1   | 6.0   | 10.2  | 3.1   | 1.1   | 0.2   | 0.0  | (17) |
| (18) |                                                                   | PrecStd     | 11.8  | 0.9   | 0.3   | 1.4   | 2.4   | 3.2   | 4.2   | 3.4   | 4.1   | 4.0   | 3.1   | 1.9   | 1.5  | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 86.6  | 92.6  | 96.9  | 101.7 | 103.1 | 96.3  | 95.1  | 91.7  | 92.2  | 91.4  | 90.4  | 86.7 | (19) |
| (20) |                                                                   | MCWB        | 69.9  | 69.0  | 71.7  | 74.8  | 76.6  | 78.8  | 78.0  | 79.3  | 79.4  | 77.7  | 76.7  | 72.8  | (20) |      |
| (21) |                                                                   | 2%          | DB    | 84.3  | 90.2  | 94.6  | 98.9  | 98.0  | 93.4  | 92.0  | 89.8  | 90.6  | 89.7  | 88.1  | 84.5 | (21) |
| (22) |                                                                   | MCWB        | 69.4  | 69.5  | 71.7  | 75.0  | 77.3  | 78.5  | 78.4  | 79.1  | 79.1  | 77.3  | 75.2  | 71.7  | (22) |      |
| (23) |                                                                   | 5%          | DB    | 82.6  | 88.1  | 92.5  | 96.2  | 94.7  | 91.1  | 89.8  | 88.1  | 88.9  | 88.1  | 86.5  | 82.7 | (23) |
| (24) |                                                                   | MCWB        | 69.1  | 69.3  | 71.3  | 75.6  | 77.9  | 78.7  | 78.7  | 78.8  | 78.5  | 76.8  | 74.4  | 70.6  | (24) |      |
| (25) |                                                                   | 10%         | DB    | 80.4  | 85.7  | 90.2  | 93.6  | 91.8  | 89.2  | 87.7  | 86.4  | 87.1  | 86.4  | 84.5  | 80.2 | (25) |
| (26) |                                                                   | MCWB        | 68.3  | 68.7  | 71.5  | 75.5  | 77.9  | 78.3  | 78.4  | 78.4  | 78.0  | 76.4  | 73.7  | 69.4  | (26) |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 74.3  | 75.1  | 77.6  | 80.6  | 81.7  | 81.5  | 81.6  | 80.9  | 81.6  | 79.7  | 78.2  | 75.4 | (27) |
| (28) |                                                                   | MCDWB       | 81.9  | 85.5  | 88.2  | 90.4  | 91.3  | 89.8  | 88.3  | 87.8  | 88.9  | 87.5  | 87.2  | 83.4  | (28) |      |
| (29) |                                                                   | 2%          | WB    | 72.0  | 73.7  | 76.0  | 79.0  | 80.2  | 80.3  | 80.3  | 80.0  | 80.1  | 78.7  | 77.1  | 73.7 | (29) |
| (30) |                                                                   | MCDWB       | 79.9  | 84.1  | 87.6  | 90.9  | 90.8  | 88.8  | 87.8  | 86.9  | 87.9  | 86.6  | 85.4  | 80.9  | (30) |      |
| (31) |                                                                   | 5%          | WB    | 70.6  | 72.0  | 74.6  | 78.0  | 79.2  | 79.5  | 79.4  | 79.4  | 79.4  | 77.9  | 76.1  | 72.4 | (31) |
| (32) |                                                                   | MCDWB       | 78.5  | 82.0  | 86.3  | 90.1  | 89.6  | 87.8  | 86.7  | 86.1  | 86.8  | 85.6  | 83.8  | 79.3  | (32) |      |
| (33) |                                                                   | 10%         | WB    | 69.5  | 70.5  | 73.2  | 76.9  | 78.4  | 78.8  | 78.7  | 78.8  | 78.7  | 77.2  | 74.9  | 70.9 | (33) |
| (34) |                                                                   | MCDWB       | 77.4  | 80.6  | 84.5  | 89.0  | 88.3  | 87.0  | 85.7  | 85.3  | 85.6  | 84.5  | 82.0  | 77.6  | (34) |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 22.7  | 29.1  | 27.5  | 23.5  | 18.0  | 14.0  | 12.2  | 11.9  | 13.4  | 15.1  | 18.9  | 20.9  | (35) |      |
| (36) |                                                                   | 5% DB       | MCDBR | 25.9  | 31.4  | 31.1  | 28.2  | 22.9  | 17.2  | 15.9  | 14.6  | 16.1  | 17.9  | 20.6  | 23.0 | (36) |
| (37) |                                                                   | MCWBR       | 11.8  | 12.4  | 10.8  | 8.4   | 6.9   | 5.4   | 5.2   | 5.5   | 6.2   | 7.0   | 8.6   | 10.6  | (37) |      |
| (38) |                                                                   | 5% WB       | MCDBR | 21.4  | 26.8  | 26.7  | 24.0  | 19.2  | 15.0  | 13.8  | 13.4  | 14.9  | 16.1  | 17.8  | 19.5 | (38) |
| (39) |                                                                   | MCWBR       | 10.5  | 11.2  | 10.9  | 9.3   | 7.3   | 5.6   | 5.6   | 5.7   | 6.5   | 6.9   | 8.0   | 9.4   | (39) |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.418 | 0.499 | 0.697 | 0.665 | 0.489 | 0.437 | 0.431 | 0.453 | 0.470 | 0.493 | 0.417 | 0.412 | (40) |      |
| (41) |                                                                   | taud        | 2.238 | 2.011 | 1.573 | 1.696 | 2.210 | 2.389 | 2.396 | 2.315 | 2.240 | 2.138 | 2.307 | 2.280 | (41) |      |
| (42) |                                                                   | Ebn at Noon | 270   | 253   | 210   | 220   | 259   | 271   | 273   | 269   | 264   | 252   | 269   | 268   | (42) |      |
| (43) |                                                                   | Edh at Noon | 41    | 55    | 88    | 78    | 46    | 38    | 38    | 42    | 44    | 48    | 38    | 38    | (43) |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 1423  | 1700  | 1750  | 1885  | 1705  | 1499  | 1362  | 1366  | 1436  | 1395  | 1420  | 1316  | (44) |      |
| (45) |                                                                   | RadStd      | 78    | 81    | 100   | 115   | 151   | 103   | 100   | 85    | 63    | 67    | 105   | 92    | (45) |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (12 neighbors) | +0.56 | N/A     | N/A    | N/A     | N/A   | +0.31 | N/A         | N/A   | +270  | +268  |

Nomenclature: See separate page